
Decoding Visual Semantics from EEG Signals: Classification and Caption Retrieval

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Abstract

This project investigates decoding visual semantics from EEG signals via three stages: EEG-based image classification, zero-shot image-caption retrieval with CLIP, and EEG-to-caption retrieval via CLIP semantic space alignment. We systematically evaluate classification architectures, preprocessing strategies, and temporal ablations to identify robust EEG representations, then extend the strongest encoder toward multimodal retrieval. CLIP zero-shot retrieval establishes an upper bound, achieving 27.7% instance-level and 96.7% class-aware Recall@1. Our best classification model achieves 9.73% validation and 9.06% test accuracy, well above the 5% random baseline. EEG-to-caption retrieval remains substantially harder, with limited gains even under partial CLIP fine-tuning, highlighting the difficulty of aligning noisy neural representations with fine-grained semantic spaces. These findings confirm that EEG signals contain partially decodable visual semantic information, but robust multimodal decoding requires stronger architectural design, preprocessing, and cross-modal representation learning.

1 Motivation and Objectives

Understanding how the human brain encodes visual information is a fundamental challenge in neuroscience and AI. Electroencephalography (EEG) offers a non-invasive, high-temporal-resolution modality for measuring neural activity, but EEG signals are inherently noisy, high-dimensional, and highly variable across subjects and sessions. Recent advances in multimodal representation learning, particularly CLIP [3], raise the question of whether neural signals can similarly be mapped into semantically meaningful embedding spaces.

We investigate this through a three-stage framework: (1) EEG-based image classification to assess coarse-grained visual category decoding and identify robust encoder architectures; (2) zero-shot CLIP image-to-caption retrieval to establish an upper bound for the pretrained semantic space; and (3) EEG-to-caption retrieval by aligning learned EEG representations with CLIP’s semantic space. Classification is evaluated via top-1 accuracy, and retrieval via Recall@K, MAP, BERTScore [4], and CLIPScore.

2 Related Work and Background

2.1 Background

EEG signals provide high temporal resolution measurements of neural activity, but their low signal-to-noise ratio and substantial cross-subject and cross-session variability make robust semantic decoding challenging. Prior work has shown that compact architectures such as EEGNet can effectively capture meaningful EEG representations for classification tasks by incorporating domain-specific temporal and spatial inductive biases [2], making EEGNet a strong foundation for visual decoding.

This project also leverages CLIP as a pretrained multimodal semantic embedding space shared between images and text [3]. Within this space, semantically related visual and textual representations are closely aligned, enabling similarity-based retrieval. EEG-to-caption retrieval can therefore be formulated as a ranking problem, where learned EEG representations are projected into CLIP’s semantic space and evaluated using Recall@K metrics.

2.2 Literature Review

Prior research has demonstrated that EEG signals contain partially decodable visual category information under supervised classification settings, with architectures such as EEGNet achieving strong performance by explicitly modeling neural signal structure [2]. However, most existing EEG decoding work focuses on coarse-grained classification rather than higher-level semantic representation.

In parallel, multimodal representation learning has significantly advanced semantic alignment across visual and textual modalities. CLIP’s large-scale contrastive framework enables robust shared semantic spaces for zero-shot classification and retrieval [3], while earlier retrieval frameworks established the effectiveness of contrastive ranking objectives for semantic correspondence learning [1]. BERTScore provides a learned evaluation metric for text generation quality by computing token-level cosine similarities between contextual embeddings [4].

Building on these developments, this project investigates whether robust EEG encoders can be systematically optimized and extended into multimodal semantic retrieval frameworks, bridging neural activity with higher-level semantic understanding beyond traditional category classification.

3 Methodology

3.1 Model Description

We evaluate a progressively expanding set of architectures to assess how architectural inductive bias, preprocessing, and multimodal alignment affect EEG semantic decoding performance across three stages: Task 1 EEG-based image classification, Task 2A zero-shot image-caption retrieval with pretrained CLIP, and Task 2B EEG-to-caption retrieval.

For Task 1, all models take as input EEG tensors of shape (122, 500) and predict one of 20 object categories. We begin with three baseline architectures of increasing complexity: an MLP baseline, a temporal CNN baseline, and EEGNet. We then extend this framework through CNN + Transformer variants, subject-aware models, multiscale EEGNet variants, and a final EEGNet residual encoder designed to produce reusable EEG embeddings for downstream multimodal alignment.

For Task 2A, we use pretrained CLIP ViT-B/32 to encode images and captions from the test split into a shared embedding space and evaluate zero-shot retrieval performance, establishing an upper bound for the semantic space before introducing EEG signals. For Task 2B, we project learned EEG embeddings into CLIP’s pretrained semantic embedding space and evaluate caption retrieval under frozen, LoRA-based, and partial fine-tuning strategies.

Figure 1 summarizes the full experimental framework, while Table 1 provides the initial baseline architecture summary.

3.2 Dataset

We use the course-provided low-speed EEG-image-caption paired dataset. Each trial is represented as an EEG matrix of shape (122, 500), where 122 denotes the number of EEG channels and 500

denotes the number of temporal samples. The classification task uses 20 object categories, while the retrieval task uses the paired captions. The final protocol contains 13 subjects and 26,000 total trials, with 15,600 training, 5,200 validation, and 5,200 test samples. We use a session-based split in which three sessions per subject are used for training, one for validation, and one for testing, evaluating cross-session generalization rather than within-session memorization.

For Task 2A, we evaluate retrieval over the 4,046 unique images in the test split, each paired with one canonical caption and one of 20 object categories. Deduplication by image name avoids trivial inflation from each image being viewed multiple times across subjects, sessions, and trials. The primary challenges across all tasks include low signal-to-noise ratio, high dimensionality, and substantial cross-subject and cross-session variability.

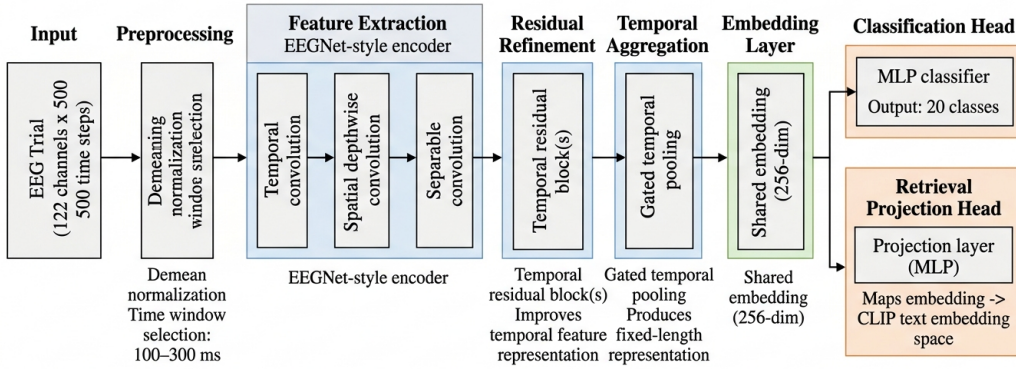


Figure 1: Overall project framework. Task 1 establishes robust EEG encoders through supervised classification and preprocessing optimization. Task 2A evaluates CLIP zero-shot image-caption retrieval as an upper bound, and Task 2B extends the EEG encoder into CLIP-aligned multimodal retrieval.

Table 1: Initial baseline model architecture summary.

Component	Description	Output Shape
Input	Raw EEG signal	(122, 500)
<i>MLP Baseline</i>		
Flatten	Vectorized input	(61000)
FC Layers	Hidden layers (512, 256) with ReLU and dropout	(256)
Classifier	Linear layer	(20)
<i>Temporal CNN Baseline</i>		
Conv1d Layers	Temporal convolutions with pooling	(256, 125)
Global Avg Pool	Temporal aggregation	(256)
Classifier	Linear layer	(20)
<i>EEGNet Baseline</i>		
Block 1	Temporal + depthwise spatial convolution	(32, 1, 125)
Block 2	Separable convolution	(32, 1, 15)
Flatten + Classifier	Linear layer	(20)

3.3 Evaluation Metrics

We evaluate performance on both classification and retrieval tasks. For classification, we use top-1 accuracy:

$$\text{Acc@1} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\hat{y}_i = y_i], \quad (1)$$

where N is the number of test samples, y_i is the ground-truth class label, and $\hat{y}_i = \arg \max_c p(y = c|x_i)$ is the predicted class. Since the dataset contains 20 balanced classes, random chance performance is $1/20 = 5\%$.

For retrieval, we use Recall@ K , which measures whether the matched item appears within the top K retrieved candidates for a given query:

$$\text{Recall@K} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\text{rank}_i \leq K], \quad (2)$$

where rank_i denotes the rank of the ground-truth match for the i -th query among all candidates. We report both instance-level Recall@ K , where only the exact paired item counts as correct, and class-aware Recall@ K , where any item from the same category counts as correct.

We additionally report MAP at caption-level and class-aware levels, BERTScore [4] for semantic similarity between retrieved and ground-truth captions, and CLIPScore for cosine similarity between image and caption embeddings within CLIP’s semantic space.

3.4 Loss Functions

For classification, we use cross-entropy loss with label smoothing. Given an EEG input $x \in \mathbb{R}^{C \times T}$, where $C = 122$ and T is the time-window length, the classifier produces logits $z = f_\theta(x) \in \mathbb{R}^{20}$. The predicted probability for class c is

$$p(y = c|x) = \frac{\exp(z_c)}{\sum_{j=1}^{20} \exp(z_j)}. \quad (3)$$

The classification loss is

$$\mathcal{L}_{cls} = - \sum_{c=1}^{20} q_c \log p(y = c|x), \quad q_c = \begin{cases} 1 - \epsilon + \frac{\epsilon}{20}, & c = y, \\ \frac{\epsilon}{20}, & c \neq y. \end{cases} \quad (4)$$

where y is the ground-truth label and ϵ is the label smoothing factor.

For retrieval, we use a contrastive alignment objective between EEG embeddings e_i and caption embeddings t_i for the i -th matched pair in a mini-batch of size N . After ℓ_2 normalization, the temperature-scaled similarity is

$$s_{ij} = \frac{\hat{e}_i^\top \hat{t}_j}{\tau}, \quad \hat{e}_i = \frac{e_i}{\|e_i\|_2}, \quad \hat{t}_i = \frac{t_i}{\|t_i\|_2}, \quad (5)$$

where τ is a temperature parameter. The symmetric contrastive retrieval loss is

$$\mathcal{L}_{ret} = \frac{1}{2} \left[-\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s_{ii})}{\sum_{j=1}^N \exp(s_{ij})} - \frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s_{ii})}{\sum_{j=1}^N \exp(s_{ji})} \right]. \quad (6)$$

When classification and retrieval are jointly optimized, the overall objective is

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda \mathcal{L}_{ret}, \quad (7)$$

where λ controls the contribution of the retrieval alignment objective.

3.5 Experimental Setup

For Task 1, models are trained for 30 epochs with AdamW using batch size 64 and a fixed random seed of 42. For the strongest final model, we additionally apply cosine learning-rate scheduling and label smoothing of 0.05, with checkpoints selected based on validation accuracy.

We systematically study two preprocessing design choices given their strong influence on EEG decoding performance: normalization strategy and temporal window selection. For normalization, we compare per-trial z-score normalization, per-channel train-set normalization, per-subject per-channel normalization, and demean-only preprocessing. For temporal windowing, we compare the full 0:500 signal with three shorter windows: 0:300, 50:350, and 100:300.

For Task 2A, we evaluate zero-shot retrieval using pretrained CLIP ViT-B/32 as the primary backbone and ViT-B/16 as an ablation. Retrieval is performed via cosine similarity ranking over the full $N \times N$ similarity matrix in both image-to-caption and caption-to-image directions. For Task 2B, we compare frozen CLIP alignment, LoRA fine-tuning, and partial fine-tuning, training for 10 epochs with batch size 64 and selecting checkpoints based on validation loss under the symmetric contrastive objective.

All experiments are conducted under consistent reproducible protocols to ensure reliable comparison across architectures, preprocessing methods, and retrieval strategies.

4 Baseline and Extensions

4.1 Initial Baseline Models

We established three baseline architectures for Task 1 EEG classification: an MLP baseline, a temporal CNN baseline, and EEGNet. The MLP served as the simplest reference point by treating EEG trials as flattened feature vectors, while the temporal CNN introduced localized temporal convolutions over EEG sequences. EEGNet provided the strongest baseline through EEG-specific temporal and spatial convolutions, offering substantially stronger inductive bias for neural signal decoding, establishing that domain-specific architectural design was significantly more important than generic model complexity.

4.2 Advanced Classification Extensions

Building on the baseline results, we explored CNN + Transformer variants, subject-aware architectures, multiscale EEGNet models, and residual EEGNet refinements. Transformer-based and subject-aware models were designed to improve long-range temporal modeling and subject generalization, but generally failed to produce consistent reproducible gains despite increased complexity. Our strongest extension was the EEGNet residual encoder, which preserved EEGNet’s effective inductive bias while introducing residual refinement, gated pooling, and reusable embedding outputs. Notably, preprocessing and optimization extensions — including normalization ablations, temporal windowing, and improved scheduling — frequently contributed more to reproducible gains than architectural complexity alone.

4.3 Retrieval Framework Extensions

For Task 2B, we extended the strongest Task 1 EEG encoder into a CLIP-aligned retrieval framework for EEG-to-caption semantic matching, evaluating frozen CLIP alignment, LoRA fine-tuning, and partial fine-tuning. Partial fine-tuning consistently produced the strongest retrieval performance, suggesting that broader semantic adaptation was necessary for effective EEG alignment. However, retrieval remained substantially more difficult than classification, highlighting the representational gap between coarse-grained visual decoding and fine-grained multimodal semantic alignment. Overall, EEG-specific inductive bias, preprocessing quality, and representation design proved more influential than raw architectural complexity in shaping the strongest final framework.

5 Results and Analysis

5.1 Baseline Classification Results

Table 2 summarizes the performance of the three initial baseline models. EEGNet achieved the strongest validation accuracy at 11.04%, approximately twice the random chance level of 5%, while the temporal CNN and MLP baselines achieved only 6.81% and 6.56%, respectively, remaining only marginally above chance.

Table 2: Classification results for the three baseline models over 30 epochs of training.

Model	Best Val Acc (%)	Best Epoch	Final Train Acc (%)	Final Val Acc (%)
MLP Baseline	6.56	29	17.0	6.19
Temporal CNN	6.81	16	40.2	5.98
EEGNet	11.04	23	19.8	9.50

These results establish that EEG-specific architectural inductive bias substantially improves semantic decoding performance, with EEGNet’s superior performance suggesting that explicit temporal-spatial modeling is essential for extracting meaningful discriminative structure from noisy EEG signals.

5.2 Training Dynamics

Figure 2 illustrates the training and validation accuracy curves for all three baseline models. Across architectures, training accuracy improved steadily while validation performance plateaued or declined, indicating overfitting as the dominant limitation. The temporal CNN exhibited the most severe overfitting, with training accuracy reaching 40.2% while validation performance declined to near chance. The MLP showed lower capacity but similarly failed to generalize effectively. EEGNet demonstrated the strongest balance between train and validation performance, achieving superior validation accuracy despite substantially lower training accuracy than the CNN, suggesting that its parameter-efficient design improved generalization rather than simple memorization.

5.3 Advanced Architecture Comparisons

To improve upon the initial baselines, we explored CNN + Transformer variants, subject-aware models, multiscale EEGNet extensions, and residual EEGNet refinements. Table 3 summarizes the strongest reproducible Task 1 models beyond the original baselines.

Table 3: Advanced Task 1 classification model comparisons under reproducible settings.

Model	Best Val Acc (%)	Test Acc (%)
Shared CNN + Transformer	6.06	—
Subject-Aware CNN + Transformer	5.87	—
Subject-Embedding CNN + Transformer	5.79	—
Multiscale EEGNet	8.29	7.29
Subject-Conditioned Multiscale EEGNet	7.46	7.17
EEGNet Residual Encoder + MLP Head	9.73	9.06

Transformer-based and subject-aware architectures did not consistently outperform simpler baselines despite increased representational complexity. In contrast, EEGNet-derived architectures remained the strongest model family, reinforcing that preserving EEG-specific inductive bias was more beneficial than increasing generic model capacity. The final residual EEGNet model produced the strongest reproducible Task 1 performance by combining refined architecture with reusable embedding-oriented design. Although an earlier EEGNet-based variant achieved the strongest exploratory result, that configuration was not reliably reproducible after the code and protocol were frozen. Since Task 2 requires a reusable EEG embedding rather than a classification-specific head, we selected the residual EEGNet encoder as the downstream backbone, preserving EEGNet’s favorable inductive bias while providing a more transfer-ready representation for multimodal alignment.

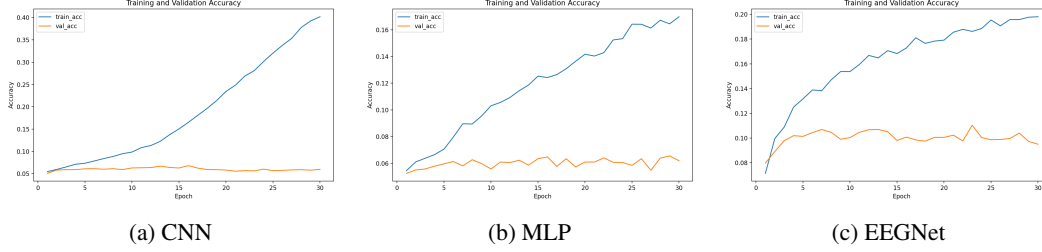


Figure 2: Training and validation accuracy curves for the three baseline models. EEGNet achieves the strongest validation performance, while all models exhibit substantial overfitting.

5.4 Preprocessing and Temporal Ablation Studies

Table 4 summarizes the main preprocessing trends within the EEGNet-based model family. Demean-only preprocessing consistently outperformed z-score normalization, and restricting the signal to the middle portion of the trial further improved validation accuracy, with the strongest setting being demean-only preprocessing with the 100:300 temporal window.

Table 4: EEGNet baseline and residual encoder comparisons under different preprocessing settings and time windows.

Setting	Normalization	Time Window	Best Val Acc (%)
EEGNet baseline	z-score per trial	0:500	7.46
EEGNet baseline	demean only	0:500	8.46
EEGNet baseline	demean only	0:300	8.96
EEGNet baseline	demean only	100:300	9.46
EEGNet residual encoder + MLP head	demean only	100:300	9.73

These findings suggest that aggressive normalization may suppress informative signal magnitude, while broader temporal windows may introduce unnecessary neural noise. Effective preprocessing therefore proved to be a major factor in maximizing usable semantic information from EEG data.

5.5 Final Task 1 Model Performance

Under the strongest reproducible configuration, the EEGNet residual encoder + MLP head achieved 9.73% best validation accuracy and 9.06% held-out test accuracy using demean-only preprocessing and the 100–300 temporal window. As shown in Figure 3, persistent train–validation divergence remained evident, indicating that overfitting and limited session generalization remained substantial bottlenecks even under the strongest final configuration.

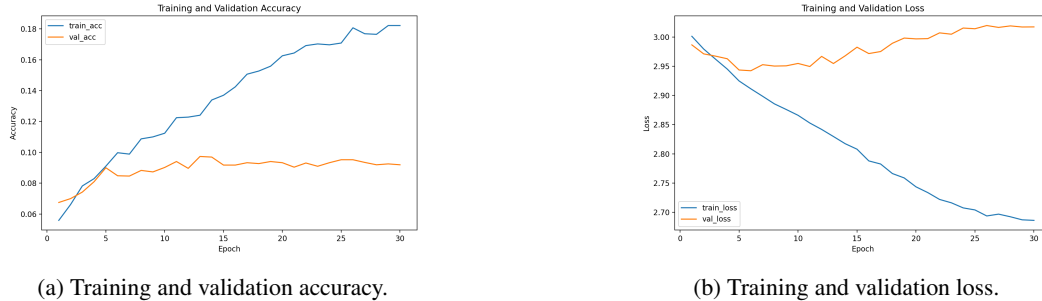


Figure 3: Training and validation dynamics for the final EEGNet residual encoder + MLP head model. Although performance improved, persistent train–validation divergence indicates that overfitting and limited session generalization remain dominant challenges.

Figure 4 further illustrates held-out test performance through the final confusion matrix. While some categories showed partial discriminability, predictions remained broadly distributed across classes, indicating that semantic separability remains limited.

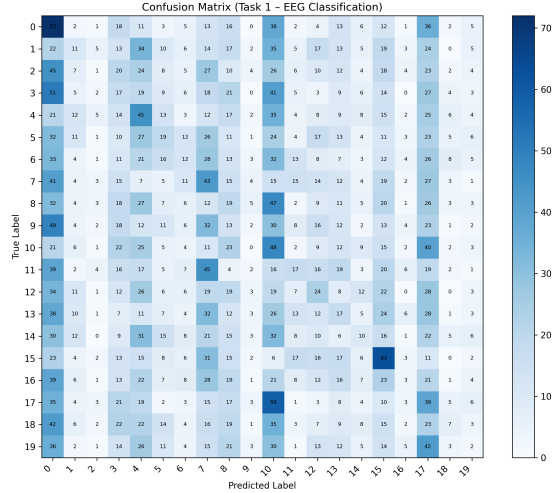


Figure 4: Confusion matrix for the final Task 1 classification model on the held-out test set. Broad inter-class confusion highlights the difficulty of robust EEG semantic decoding.

5.6 Task 2A: Zero-Shot Image-Caption Retrieval with CLIP

Before introducing EEG embeddings into the retrieval pipeline, we evaluated CLIP’s zero-shot image-caption retrieval performance on the test split as an approximate upper bound for Task 2B. We used CLIP ViT-B/32 as the primary backbone and evaluated 4,046 unique test images after deduplicating the 5,200 test trials by image name, with retrieval performed via cosine similarity over the full 4046×4046 image-caption similarity matrix.

Table 5: Task 2A zero-shot retrieval performance using CLIP ViT-B/32 on 4,046 unique test images. Instance-level retrieval requires the exact paired item, while class-aware retrieval accepts any item from the same object category.

Direction	Type	R@1	R@3	R@5	R@10
Image → Caption	Instance	0.277	0.448	0.536	0.652
Image → Caption	Class-aware	0.967	0.981	0.986	0.991
Caption → Image	Instance	0.235	0.376	0.453	0.573
Caption → Image	Class-aware	0.969	0.993	0.995	0.997

Table 5 shows a large gap between instance-level and class-aware retrieval. For image-to-caption retrieval, instance-level Recall@1 reached 27.7% while class-aware Recall@1 reached 96.7%, indicating that CLIP’s semantic space strongly organizes samples by category, but exact instance matching remains harder as many captions within the same category are semantically similar.

Table 6 further supports this interpretation: caption-level MAP was 0.399 while class-aware MAP was 0.831, confirming that CLIP is more reliable at category-level retrieval than exact instance matching. Top-1 retrieved captions remained semantically close to the ground truth, with mean BERTScore F1 of 0.933 and a 100% success rate under the $F1 > 0.7$ threshold.

Figure 5 visualizes these trends. The BERTScore histogram confirms all top-1 retrieved captions exceed the $F1 > 0.7$ threshold, and the CLIPScore distributions show strong separation between matched and mismatched pairs. The per-class MAP plot shows that visually distinctive categories such as flower, tvmonitor, horse, and aeroplane achieve the strongest retrieval performance, while visually diverse categories such as person and car are more difficult.

Table 6: Additional Task 2A metrics for CLIP ViT-B/32.

Metric	Value
Caption-level MAP	0.399
Class-aware MAP	0.831
BERTScore F1, mean / std / median	0.933 / 0.049 / 0.923
BERTScore success rate, F1 > 0.7	100%
BLEU / ROUGE-L / METEOR, mean	0.345 / 0.572 / 0.520
Matched-pair CLIPScore, mean / std	0.302 / 0.031
Random mismatched CLIPScore, mean / std	0.144 / 0.042
Different-class mismatched CLIPScore, mean / std	0.140 / 0.036
KS statistic, matched vs. random	0.939
KS statistic, matched vs. different-class	0.968

As an ablation, we compared ViT-B/32 against ViT-B/16. ViT-B/16 produced slightly stronger Recall@1 and MAP results, likely due to its smaller patch size providing higher-resolution visual features, but the gains were small (Table 7).

Table 7: Task 2A CLIP backbone ablation.

Metric	ViT-B/32	ViT-B/16
Image → Text R@1, instance	0.277	0.294
Image → Text R@5, instance	0.536	0.531
Image → Text R@1, class-aware	0.967	0.974
Caption-level MAP	0.399	0.410
Class-aware MAP	0.831	0.839

Overall, Task 2A demonstrates that CLIP’s pretrained semantic space is highly effective for this dataset when clean image embeddings are available. This makes the later drop in Task 2B performance more informative: the bottleneck is not the caption space itself, but the difficulty of mapping noisy EEG signals into that semantic space.

5.7 Task 2B: EEG-to-Caption Retrieval Results

Task 2B extended the strongest EEG encoder into multimodal semantic retrieval via alignment with CLIP’s pretrained semantic embedding space. Table 8 summarizes retrieval performance across all adaptation strategies.

Table 8: Task 2B EEG-to-caption retrieval performance across CLIP adaptation strategies. The random baseline is computed as $K/5200$ because retrieval is evaluated over 5,200 test trials.

Method	Recall@1	Recall@5	Recall@10
Random Baseline	0.00019	0.00096	0.00192
Frozen CLIP Alignment	0.00038	0.00154	0.00327
LoRA Fine-Tuning	0.00058	0.00212	0.00327
Partial Fine-Tuning	0.00096	0.00250	0.00500

Partial fine-tuning produced the strongest retrieval metrics, suggesting broader semantic adaptation improves EEG alignment, though performance remained extremely limited overall. Comparing Task 2A and 2B highlights the bottleneck: CLIP achieved 27.7% Recall@1 with clean image embeddings, while the strongest Task 2B model reached only 0.096% with EEG-derived embeddings. Though still above the 0.019% random baseline, this gap confirms that the primary difficulty is learning EEG representations that preserve sufficient semantic information to align with the pretrained CLIP space.

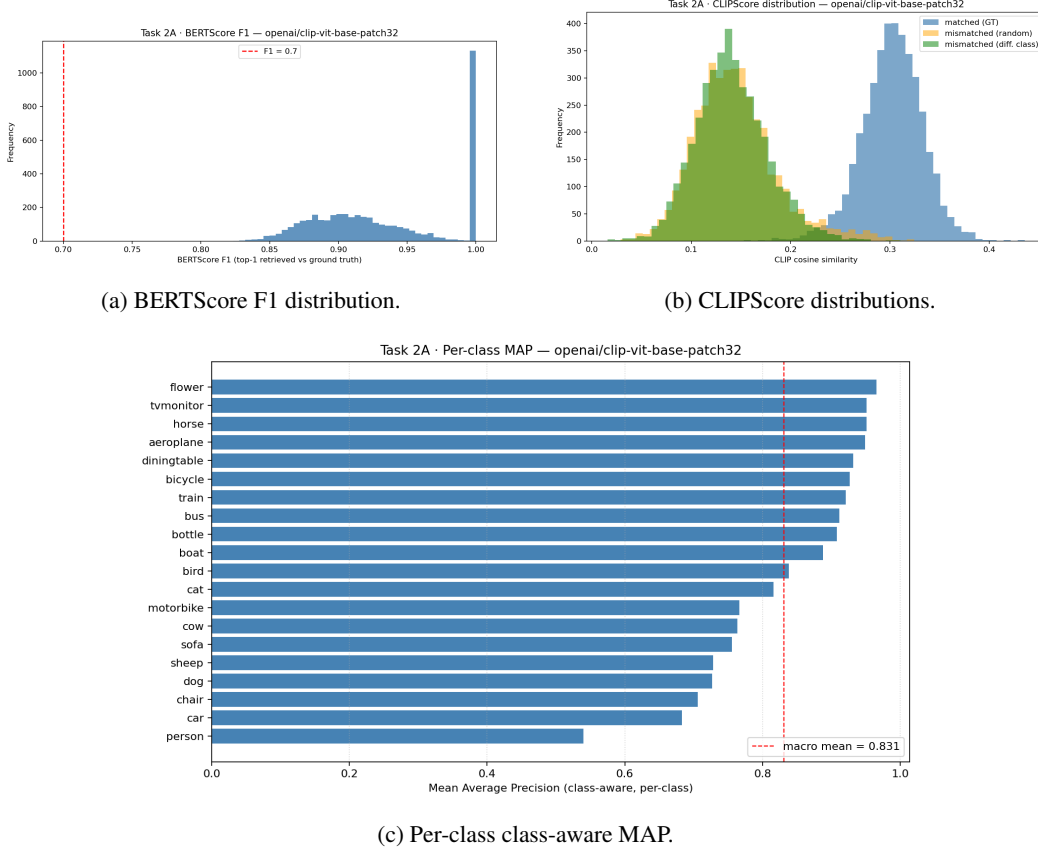


Figure 5: Task 2A retrieval analysis using CLIP ViT-B/32. The BERTScore distribution shows that top-1 retrieved captions are consistently semantically close to the ground truth. The CLIPScore distribution shows strong separation between matched and mismatched image-caption pairs, with limited overlap. Per-class MAP shows that visually distinctive classes such as flower, tvmonitor, horse, and aeroplane are easier, while more diverse classes such as person and car are harder.

5.8 Overall Analysis

Across both tasks, EEG-specific architectural inductive bias was a critical determinant of performance, with EEGNet-derived models consistently outperforming generic MLP, CNN, and Transformer architectures. Preprocessing and temporal window selection proved equally important, often producing larger gains than architectural modifications alone. While Task 2A confirmed that CLIP provides a strong semantic retrieval space at the category level, Task 2B revealed that replacing clean image embeddings with EEG-derived embeddings causes a dramatic performance drop, indicating that the primary bottleneck is learning sufficiently semantic EEG representations rather than the retrieval framework itself. Overfitting and cross-session variability remained dominant limitations throughout, highlighting that robust multimodal EEG semantic retrieval remains an open challenge.

6 Discussion and Future Work

The results demonstrate meaningful progress toward EEG-based semantic decoding while highlighting substantial remaining limitations. Persistent overfitting across nearly all architectures indicates that current models struggle to learn sufficiently stable EEG representations under noise and recording variability, with classification accuracy remaining well below practical deployment thresholds. A notable finding is that preprocessing choices, particularly temporal window selection, contributed as much or more to performance gains than several architectural changes, and should be treated as a primary design consideration for this dataset.

Future work should focus on three directions: improving Task 1 generalization through stronger scheduling, augmentation, and regularization; developing subject-adaptive and domain-generalization methods to address cross-session variability; and strengthening multimodal representation learning through stronger contrastive objectives, improved embedding architectures, and alternative alignment frameworks. Collectively, these directions may help bridge the gap between coarse-grained EEG classification and practical semantic-level neural decoding.

7 Conclusion

This project established a progressively refined framework for EEG-based semantic decoding, spanning baseline classification, advanced architectural extensions, preprocessing ablations, zero-shot CLIP image-caption retrieval, and multimodal EEG-to-caption retrieval. EEG-specific architectural design and focused preprocessing consistently improved classification performance, with the final EEGNet residual encoder + MLP head achieving the strongest reproducible results, confirming that EEG contains partially decodable visual semantic information.

Despite these improvements, overfitting, session variability, and limited generalization remained major constraints, and Task 2B retrieval performance remained extremely limited even under partial fine-tuning, highlighting the substantial gap between category-level classification and fine-grained semantic understanding from EEG. Overall, meaningful progress is achievable through careful architectural design, preprocessing, and multimodal modeling, but substantial challenges remain before robust, practical EEG-based semantic retrieval becomes feasible.

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Team Contributions

Jiachen Ye: Set up the GitHub repository and project workflow, implemented the data preprocessing and dataset splitting pipeline, led Task 1 and Task 2B, drafted the Abstract and Motivation and Objectives sections, and coordinated the final report revisions.

Takeshi Koey: Led Task 2A, including CLIP-based retrieval evaluation and analysis, experimented with Task 1 architectures, and contributed to the Results and Analysis section, particularly the Task 2A results and Task 2A–2B comparison.

Aadhil Mubarak Syed: Primarily responsible for verifying experimental results and reported metrics, and wrote and revised the majority of the final report, including the Literature Review, Methodology, Baseline and Extensions, Results and Analysis, and Discussion sections.

Github: <https://github.com/yjcexp/11685-Guided-Project>